Quiz 6

Graded

Student

[redacted]

Total Points

10 / 30 pts

Question 1

(no title)

3 / 10 pts

- **+ 10 pts** Correct
- **+ 5 pts** Only found 6 as an eigenvalue
- **+ 4 pts** Found lambda= 0 with multiplicity 2
- **+ 2 pts** Found lambda = 0 with multiplicity 1
- **+ 1 pt** Did not find any eigenvalues and was not on the right track
- ✔ **+ 3 pts** Incorrect but on the right track

Question 2

(no title)

5 / 10 pts

- **+ 10 pts** Correct
- **+ 8 pts** Correct but poor justification
- ✔ **+ 5 pts** Correct with no/wrong justification or incorrect for the right reason
- **+ 3 pts** Incorrect but reasoning shows understanding
- **+ 0 pts** Incorrect and reasoning shows poor understanding

Question 3

(no title)

2 / 10 pts

- **+ 5 pts** Part a) is correct
- **+ 5 pts** Part b) is correct
- **+ 3 pts** approach for part a) shows some understanding but is incorrect or only partially correct
- **+ 3 pts** approach for part b) shows some understanding but is incorrect or only partially correct
- **+ 0 pts** Click here to replace this description.
- ✔ **+ 1 pt** part a) was attempted
- ✔ **+ 1 pt** part b) was attempted

Math 54 Quiz 6

Name: [redacted] ID: ____________

Problem 1

Find all eigenvalues (with multiplicity) of the following matrix: (*Hint: There is a way to solve this problem without doing any calculations at all but you can of course do it any valid way you like.*)

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$$

conj

eliminate all (if any) dependent rows or columns $\in A^T \wedge A$,

then, suppose the pivots $\in A := (A - \lambda) = 0$ are D (diagonal)

representation

we can express A in parametric form in terms of

free variables $\forall$ show that D's diagonals are the

eigen values $\in A$

Problem 2 $\mathcal{B} \in V$ $\underset{C \leftarrow B}{P}$

(T/F) Let $\mathcal{B} = \{b_1, \ldots, b_n\}$ be a basis for a vector space V. The change-of-basis matrix from the $\underbrace{\text{standard basis to the } \mathcal{B} \text{ basis is}}$ the matrix $P_{\mathcal{B}}$ whose columns are the b_i. (Provide justification.)

$[\mathcal{E}]$ $[x]_B = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$

we know the change of basis $\underset{C \leftarrow B}{P} = [[b_1]_{C_1}, \ldots, [b_n]_{C_n}] = [\mathcal{E}]_B$

Thus this statement is true

Problem 3

a) Prove that if A is diagonalizable and B is similar to A, then B is also diagonalizable.

b) Prove that if A is invertible and A is similar to B, then B is invertible as well. (Brownie points - but no bonus points - if you also show that A^{-1} is similar to B^{-1})

a) if $\det(A) \wedge \det(B) \neq 0$, and W.T.S $A \sim B \rightarrow$

$A = PBP^{-1}$: $\quad A = PP^{-1}B$

$A = IB \Rightarrow A = B$

Thus, $A \wedge B$ are similar and there diagonalizable values are equivalent

b) if A has rank(a) for $\mathbb{R}^n \wedge A \sim B$ (proven in part a)

then both satisfy the condition that $A \wedge B$ are invertible because there is full independence, meaning:

$$A \rightarrow C_{ij}M_{ij} \rightarrow B^T \rightarrow \frac{1}{ad-bc}\begin{bmatrix} d & -(b) \\ -(c) & a \end{bmatrix}$$

and it follows that $A^{-1} \sim B^{-1}$